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Title:

Photometric periods of new X-ray selected Cataclysmic Variables

Abstract:

The proposal aims to better constrain close binary evolution and to determine the space density of accreting compact white-dwarf binaries. The proposers have embarked on a comprehensive optical identification program of newly detected cataclysmic variables (CVs) from the eROSITA all-sky survey. The first ~ 150 new CVs were identified spectroscopically. To fully characterize the new objects and to uncover their evolutionary role one needs to determine their orbital period. Here we propose to observe 22 targets in addition to the 22 targets we applied for through the regular CNTAC call. The additional targets have been spectroscopically identified as CVs through SDSS spectroscopy after the regular CNTAC deadline.

Telescope & Instrument: SPECULOOS with CCD

Time requested: 22 nights preferred, 10 nights minimum

SCIENTIFIC AIM AND RATIONALE

Cataclysmic variables (CVs) are the most common endproducts of close binary evolution (1). They consist of a white dwarf (WD) primary that accretes matter via Roche-lobe overflow (RLOF) from a low-mass companion (8). More than 90% of the systems have binary orbital periods below 4 hours and, in very general terms, they evolve under the influence of angular momentum loss (AML) through magnetic braking and gravitational radiation, from long to short orbital periods. Understanding close binary evolution is of imminent relevance to uncover pathways to SNIa and to understand the foreground signal of the planned space-based gravitational wave observatory. Yet there are fundamental unsolved problems concerning their sample properties and their evolution that shall be tackled by this proposal.

Accretion onto the WD is strongly dictated by its magnetic field. In CVs where the WD has a weak magnetic field an accretion disk forms that may extend to the surface of the WD, while in CVs with highly magnetized WDs (magnetic CVs, mCVs), known as polars and intermediate polars, there is at most a truncated disk and matter flows via accretion columns onto the magnetic poles of the WD. Observationally mCVs can be distinguished from non-magnetic systems by their emission lines profiles (symmetric disk lines for non-magnetic CVs, asymmetric lines from streams in mCVs), the strength of the He II $\lambda 4686$ emission line (similar to that of $H\beta$ in mCVs, weak or absent in non-magnetic CVs), and the occasional detection of cyclotron harmonic emission lines or Zeeman absorption lines as direct indicators of magnetism. Photometrically, most CVs show periodic variability on the binary orbital period (unless the orbital inclination is too small, which applies to a very small subset of the known population) and may show periodic variability also on the spin period of the accreting WD ($P_{\text{orb}} = P_{\text{spin}}$ for the polars, $P_{\text{spin}} \sim 0.1P_{\text{orb}}$ for the intermediate polars, flickering variability for the non-magnetic CVs.)

The evolutionary status of a given single system is written into its binary orbital period and the evolution of the class (and the sub-classes) is written into the binary orbital period distribution. Thus, the binary orbital period is *the* fundamental parameter to be determined for individual systems and complete samples to fully understand binary evolution.

There are two fundamental issues that were hampering progress in the past, one on the theoretical, one on the observational side. In the past, binary population synthesis (BPS) models (e.g. by 3) have completely neglected to take into account the effect of magnetic fields. It was shown by Belloni et al. (2), that magnetic fields will play an integral role in the evolution of compact binaries by weakening the effect of magnetic braking, and thereby slowing down the angular momentum loss (AML) in mCVs. Observationally only recently it has become clear that mCVs and non-magnetic systems have significantly different orbital period distributions (6), which implies that the strength of the magnetic field plays a significant role and forces mCVs to have different evolutionary pathways from non-magnetic systems. However, the strength of AML is not fully quantified so that a full-scale BPS that could be compared with observations is still missing.

On the observational side, sufficiently large complete samples are not yet available so that neither the space density nor the fraction of mCVs among all CVs are known. The space density has uncertainties of a factor of 2 or larger which is simply due to the small sizes of either flux- or volume-limited samples (7; 5). Pala's sample consists of 42 CVs, has an estimated completeness $>80\%$, and comprises a fraction of $\sim 36\%$ mCVs. However, a first eROSITA-selected flux-limited sample with 23 CVs contained only 2 magnetic systems (5).

The prospects to overcome the current observational limitations are excellent. All CVs are X-ray emitters (4) and this property allows to build large and complete volume-limited samples from the exploitation of the sensitive X-ray all-sky survey performed by eROSITA. From a probabilistic source match of eROSITA X-ray sources with Gaia objects a first batch of high-confidence CVs was selected. The target properties that led to their selection are described and illustrated in Fig. 2.

From the analysis of existing EFOSC2@NTT spectroscopy of 150 candidates the target association could be quantified to be better than $> 90\%$, as 133 new CVs were confirmed. eROSITA data allows to build a volume-limited sample of the various CV sub-classes out to 500 pc. The whole sample will comprise of about 1000 systems. The subset of 22 targets we propose to observe through this FastTrack proposal have been identified through SDSS spectroscopy after the regular CNTAC deadline. All the CVs in our target list are new to the literature and lack knowledge about their orbital periods. These can most easily be obtained from dedicated time-resolved photometric observations, that is proposed here. If time can be granted, this would significantly speed up our progress towards establishing a large representative sample of CVs. All the systems in our subset have equal priority for observations as no orbital period has been found for any of them.

We propose to perform time-resolved photometric observations of new confirmed CVs for which nothing more than their X-ray properties and an identification spectrum are known. Two example spectra obtained with EFOSC2 on NTT are shown in Fig. 1. The strong HeII emission line at $4686\text{\AA}$ clearly identify the object as a strong mCV candidates. The determination of the binary orbital periods of all the objects listed below is the one and only aim of this proposal. This measurement will deliver the missing piece of information to make use of the newly discovered CVs for sample studies and evolutionary considerations.

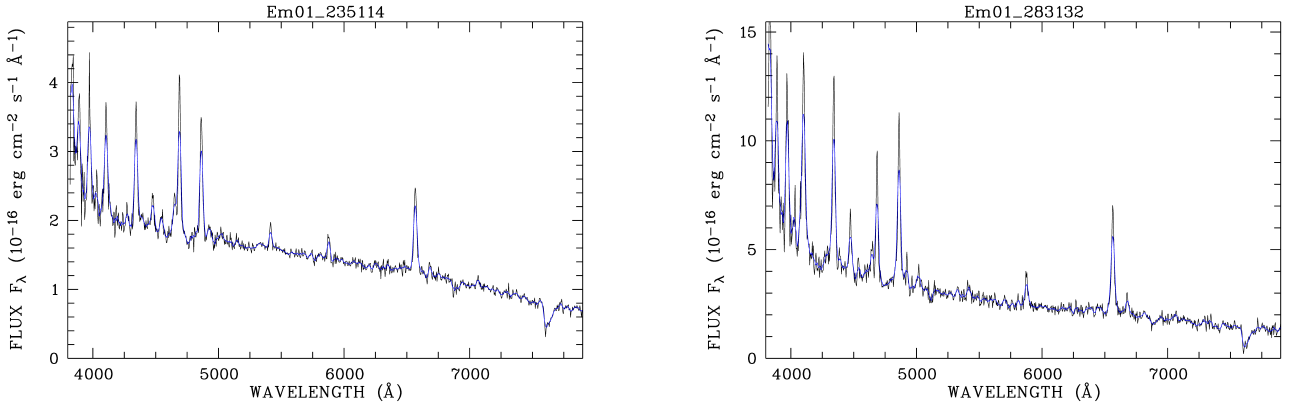


Figure 1: NTT identification spectra obtained with EFOSC2 of two of the proposed targets, showing very strong HeII $\lambda 4686$ emission lines, indicating the likely magnetic nature of the targets.

References

- [1] Belloni D., Schreiber M. R., 2023, in , Handbook of X-ray and Gamma-ray Astrophysics. Edited by Cosimo Bambi and Andrea Santangelo. p. 129, [doi:10.1007/978-981-16-4544-0_98-1](https://doi.org/10.1007/978-981-16-4544-0_98-1)
- [2] Belloni D., Schreiber M. R., Pala A. F., Gänsicke B. T., Zorotovic M., Rodrigues C. V., 2020, *MNRAS*, **491**, 5717
- [3] Goliasch J., Nelson L., 2015, *ApJ*, **809**, 80
- [4] Mukai K., 2017, *PASP*, **129**, 062001
- [5] Pala A. F., et al., 2020, *MNRAS*, **494**, 3799
- [6] Schreiber M. R., Belloni D., Schwöpe A. D., 2024, *A&A*, **682**, L7
- [7] Schwöpe A. D., 2018, *A&A*, **619**, A62
- [8] Warner B., 1995, Cataclysmic variable stars. Vol. 28

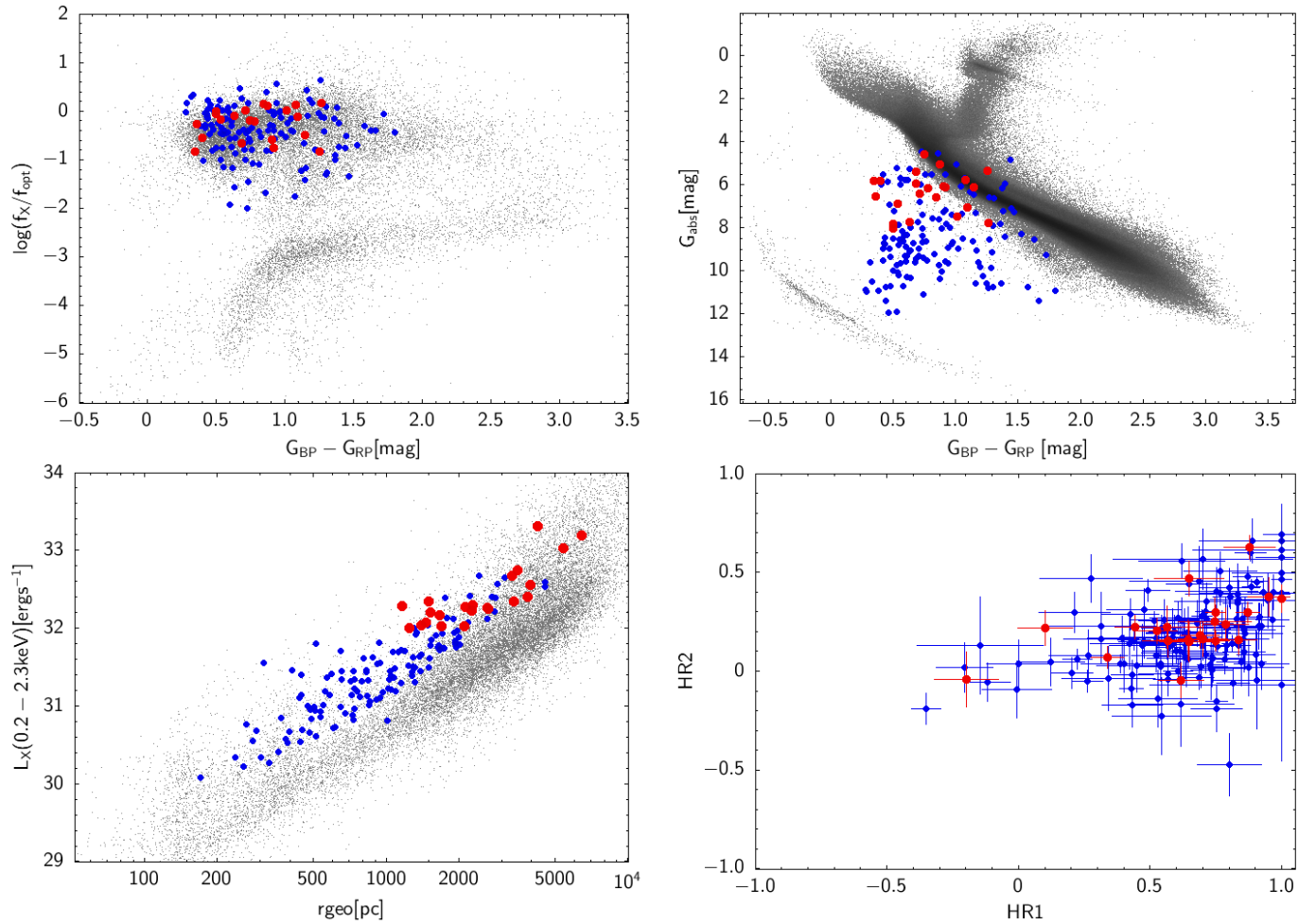


Figure 2: Diagnostic diagrams of newly identified CVs from eROSITA. Shown are an X-ray/optical color-color diagram, a color-magnitude diagram, a luminosity-distance diagram, and an X-ray color-color diagram. Shown in blue are all new CVs, those selected for this proposal are shown in red. The X-ray/optical color-color diagram shows the comparably high X-ray flux of the CVs (as high as those of AGN and 2-3 orders of magnitude higher than stars, both shown as grey background sources). The CMD illustrates the large spread in absolute magnitude thus covering all kinds of CVs (the grey background is made of non-X-ray emitting unrelated objects to outline the ZAMS and the WD sequence). The L_X -dist diagram illustrates the potential to build large complete samples out to 500 pc (grey background objects are selected from deeper eROSITA surveys under investigation in e.g. SDSS), and the X-ray hardness ratio diagram separates mCVs from non-magnetic ones through their soft X-ray spectra with $HR1 < 0.3$. A hardness ratio is defined as $HR = (H - S) / (H + S)$, with H and S being X-ray photons in two adjacent energy bands. With the given normalization, HR -values are constrained within $+1$ and -1 . An $HR1$ -value above 0.3 is compatible with just thermal emission from an accretion column or the disk-WD boundary layer. A value below 0.3 indicates the presence of a soft, blackbody-like source, as discovered from the heated photospheres in the vicinity of the accretion regions in mCVs.

TECHNICAL DESCRIPTION

For each of the targets in the list below we request time-resolved photometry during two successive nights with SPIRIT@SPECULOOS lasting at least 4 hours each. We choose the r filter for its high sensitivity. We plan to use integration times of 60 sec (for objects brighter than $G=17.5$ we will use 30 sec exposures). The observations shall be performed without interruption for the planned $\gtrsim 4$ hours per night. The first observation will give a first estimate of the binary orbital period. A second observation in the following night is requested to then determine the period with about 10% accuracy, sufficient for the later analysis in an evolutionary context.

For each target we request 2 observations, each lasting at least 4 hours. Given the duration of the nights in the observation period, we can therefore observe two targets per night, with each observation lasting at least 4 hours per target, and therefore request 22 nights of observations. **While this would significantly slow down our progress, we could continue the project with a smaller number of targets and slightly reduced observing time. However, allocating less than 10 nights (corresponding to 10 targets) would not make sense in the light of our ambitious goals.** The targets for this run form part of a larger list of targets for which time-resolved photometric observations are required to determine their orbital periods, and hence future SPECULOOS time will be requested as the project progresses.

Name	RA (deg)	DEc (deg)	G (mag)
eb01_216081.020_ML00131_002_c946	14:29:53.61	+07:32:32.34	20.227272
em01_186153.020_ML00237_002_c946	12:23:21.83	-61:47:54.24	20.846275
em01_191150.020_ML00160_002_c946	12:41:00.75	-59:30:30.90	16.20195
em01_205153.020_ML00242_005_c946	13:38:46.35	-62:35:50.93	18.471596
em01_218156.020_ML00004_002_c946	14:30:00.12	-67:14:41.84	16.697113
em01_220150.020_ML00032_002_c946	14:37:21.89	-58:47:37.23	16.688957
em01_230096.020_ML00033_002_c946	15:18:35.96	-05:48:03.24	19.329763
em01_244144.020_ML00004_002_c946	16:25:28.99	-55:20:00.25	20.713856
em01_244144.020_ML00015_002_c946	16:06:39.13	-54:32:06.59	20.730394
em01_246147.020_ML00076_005_c946	16:17:06.77	-56:27:17.75	20.903477
em01_247141.020_ML00024_006_c946	16:19:32.27	-49:44:25.34	19.945778
em01_253156.020_ML00053_003_c946	16:42:05.92	-65:53:28.49	14.003806
em01_255132.020_ML00009_001_c946	17:04:04.84	-43:05:34.28	16.31228
em01_256141.020_ML00079_002_c946	16:58:59.90	-51:23:29.26	16.534594
em01_259123.020_ML00035_005_c946	17:16:58.01	-31:40:00.26	20.945715
em01_259123.020_ML00079_005_c946	17:11:59.70	-33:35:02.73	19.143673
em01_271150.020_ML00023_002_c946	18:12:43.48	-59:23:11.19	17.249615
em01_273147.020_ML00022_003_c946	18:01:11.34	-58:07:04.37	18.318907
em01_287153.020_ML00068_002_c946	19:01:20.99	-62:28:04.22	19.75015
em01_294156.020_ML00035_001_c946	19:42:55.05	-67:27:27.21	20.100092
em01_299147.020_ML00013_001_c946	19:48:28.85	-56:29:40.33	18.254242
em01_304147.020_ML00140_001_c946	20:25:43.43	-58:23:34.12	16.340748

CURRENT STATUS OF THE PROJECT

Research performed by our group on magnetic white dwarfs and white dwarf binary evolution had a global impact. We showed that the evolution of magnetic white dwarfs in cataclysmic variables can be understood by incorporating reduced magnetic braking in the standard prescriptions for the evolution of these systems (1). We demonstrated that one of the scenarios frequently suggested for the generation of magnetic fields in white dwarfs fails to explain the observations of detached and semi-detached white dwarf binaries (2). We also published a comprehensive review on the X-ray emission of magnetic cataclysmic variables in ApJS (3) and a new idea for the magnetic field generation in white dwarfs in close binary stars in Nature Astronomy (4). Our partners are running a highly successful identification program of compact sources from the eROSITA surveys (5) which led to the discovery of the 2nd white dwarf pulsar (6) and allows to establish a large and volume limited sample of accreting white dwarf binaries. We have started a follow-up programs of these systems with the SDSS (7) and ESO telescopes. We here propose to measure the orbital period of systems discovered by eROSITA and confirmed through spectroscopy. The orbital period distribution of these systems is key to understand their evolution (8). Throughout the next four years we will combine eROSITA and SDSS discoveries of new CVs (led by Schwöpe and Schreiber) with follow-up observations (performed and analyzed by two PhD students, Jaco Brink from AIP and Sarai Zapata from UTFSM). This project plays a crucial role in the development of white dwarf binary research in the region of Valparaíso. In particular, combining detections of the European X-ray satellite (eROSITA) with follow-up observations using telescopes located in our country represents a fantastic win-win situation. At the end of October 2024 the first SPECULOOS data has been taken for this project and is currently analyzed.

Relevant publications

- [P1] D. Belloni, M. R. Schreiber, A. F. Pala, B. T. Gänsicke, M. Zorotovic, and C. V. Rodrigues, “Evidence for reduced magnetic braking in polars from binary population models,” *MNRAS*, vol. 491, pp. 5717–5731, Feb. 2020.
- [P2] D. Belloni and M. R. Schreiber, “Are white dwarf magnetic fields in close binaries generated during common-envelope evolution?,” *MNRAS*, vol. 492, pp. 1523–1529, Feb. 2020.
- [P3] D. Belloni, C. V. Rodrigues, M. R. Schreiber, M. Castro, J. E. R. Costa, T. Hayashi, I. J. Lima, G. J. M. Luna, M. Martins, A. S. Oliveira, S. G. Parsons, K. M. G. Silva, P. E. Stecchini, T. J. Stuchi, and M. Zorotovic, “Breaking the Degeneracy in Magnetic Cataclysmic Variable X-Ray Spectral Modeling Using X-Ray Light Curves,” *ApJS*, vol. 256, p. 45, Oct. 2021.
- [P4] M. R. Schreiber, D. Belloni, B. T. Gänsicke, S. G. Parsons, and M. Zorotovic, “The origin and evolution of magnetic white dwarfs in close binary stars,” *Nature Astronomy*, Apr. 2021.
- [P5] A. Schwöpe, J. Kurpas, P. Baecke, K. Knauff, L. Stütz, D. Tubín-Arenas, A. Standke, S. F. Anderson, F. Bauer, W. N. Brandt, K. Covey, S. Demasi, T. Dwelly, S. Freund, S. Friedrich, B. T. Gänsicke, C. Maitra, A. Merloni, D. Muñoz-Giraldo, A. Rodriguez, M. Salvato, K. Stassun, B. Stelzer, A. Strong, and S. Morrison, “Compact white dwarf binaries in the combined SRG/eROSITA/SDSS eFEDS survey,” *A&A*, vol. 686, p. A110, June 2024.
- [P6] A. Schwöpe, T. R. Marsh, A. Standke, I. Pelisoli, S. Potter, D. Buckley, J. Munday, and V. Dhillon, “X-ray properties of the white dwarf pulsar eRASSU J191213.9–441044,” *A&A*, vol. 674, p. L9, June 2023.
- [P7] A. Schwöpe, K. Knauff, J. Kurpas, M. Salvato, B. Stelzer, L. Stütz, and D. Tubín-Arenas, “A first systematic characterization of cataclysmic variables in SRG/eROSITA surveys,” *arXiv e-prints*, p. arXiv:2407.20903, July 2024.
- [P8] M. R. Schreiber, D. Belloni, and A. D. Schwöpe, “The cataclysmic variable orbital period gap: More evident than ever,” *A&A*, vol. 682, p. L7, Feb. 2024.